

IMAGE
ENHANCEMENT
OF LANDSCAPE
USING HSV AND
SPATIAL FILTERING

MINI PROJECT 1

OBJECTIVE:

- **ENHANCING SHADOW VISIBILITY**
- **IMPROVING CLARITY WHILE PRESERVING
NATURAL COLORS**
- **DECREASE THE OVEREXPOSURE
BRIGHTNESS OF THE SUN**

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PROBLEM STATEMENT

- FOREGROUND (PEOPLE & DECK) TOO DARK
- LOSS OF DETAIL IN SHADOW REGIONS
- SLIGHT HAZE REDUCING CLARITY
- RISK OF OVEREXPOSING THE SUN DURING
ENHANCEMENT

02\10



CONCEPT USED

```
%% SUNSET_SHADOW_VISIBLE_BALANCED_SUN.M
clear; close all; clc;

inFile = 'sunset.JPG';
img = im2double(imread(inFile));
if size(img,3)==1, img = repmat(img,[1 1 3]); end
```

Image normalization (im2double)

```
%% 1) HSV
hsvImg = rgb2hsv(img);
H = hsvImg(:,:,1);
S = hsvImg(:,:,2);
V = hsvImg(:,:,3);
```

HSV color space processing
(rgb2hsv, hsv2rgb): RGB to HSV

CONCEPT USED(CONTI.)

%% 2) Shadow Lift (same as before)

```
shadowMask = V < 0.50;
```

```
V1 = V;
```

```
V1(shadowMask) = V1(shadowMask).^0.45;
```

```
% Protect highlights
```

```
highMask = V1 > 0.85;
```

```
V1(highMask) = 0.85 + 0.15*((V1(highMask)-0.85).^1.5);
```

```
% Boost saturation in shadows
```

```
S1 = S;
```

```
S1(shadowMask) = min(max(S1(shadowMask) + 0.18,0),1);
```

Power-law (gamma) transformation
& Highlight Compression

%% 3) 🔥 NEW STEP: Reduce Sun Brightness Only

```
% Detect extremely bright region (sun area)
```

```
sunMask = V1 > 0.92;
```

```
% Apply mild compression only to sun
```

```
V1(sunMask) = V1(sunMask).^1.8; % gamma >1 darkens bright
```

```
% Slightly reduce saturation of sun to avoid glow
```

```
S1(sunMask) = S1(sunMask) * 0.85;
```

Sun Reduction

CONCEPT USED(CONTI.)

```
%% 4) Back to RGB
```

```
rgb1 = hsv2rgb(cat(3, H, S1, min(max(V1,0),1)));
```

HSV color space processing
(rgb2hsv, hsv2rgb): HSV to RGB

```
%% 5) Sharpen (same as before)
```

```
Y = rgb2gray(rgb1);
```

```
Yb = imgaussfilt(Y,1.0);
```

```
detail = Y - Yb;
```

```
amount = 0.60;
```

```
rgbOut = rgb1;
```

```
for c = 1:3
```

```
    rgbOut(:,:,c) = min(max(rgbOut(:,:,c) + amount*detail,0),1)
```

```
end
```

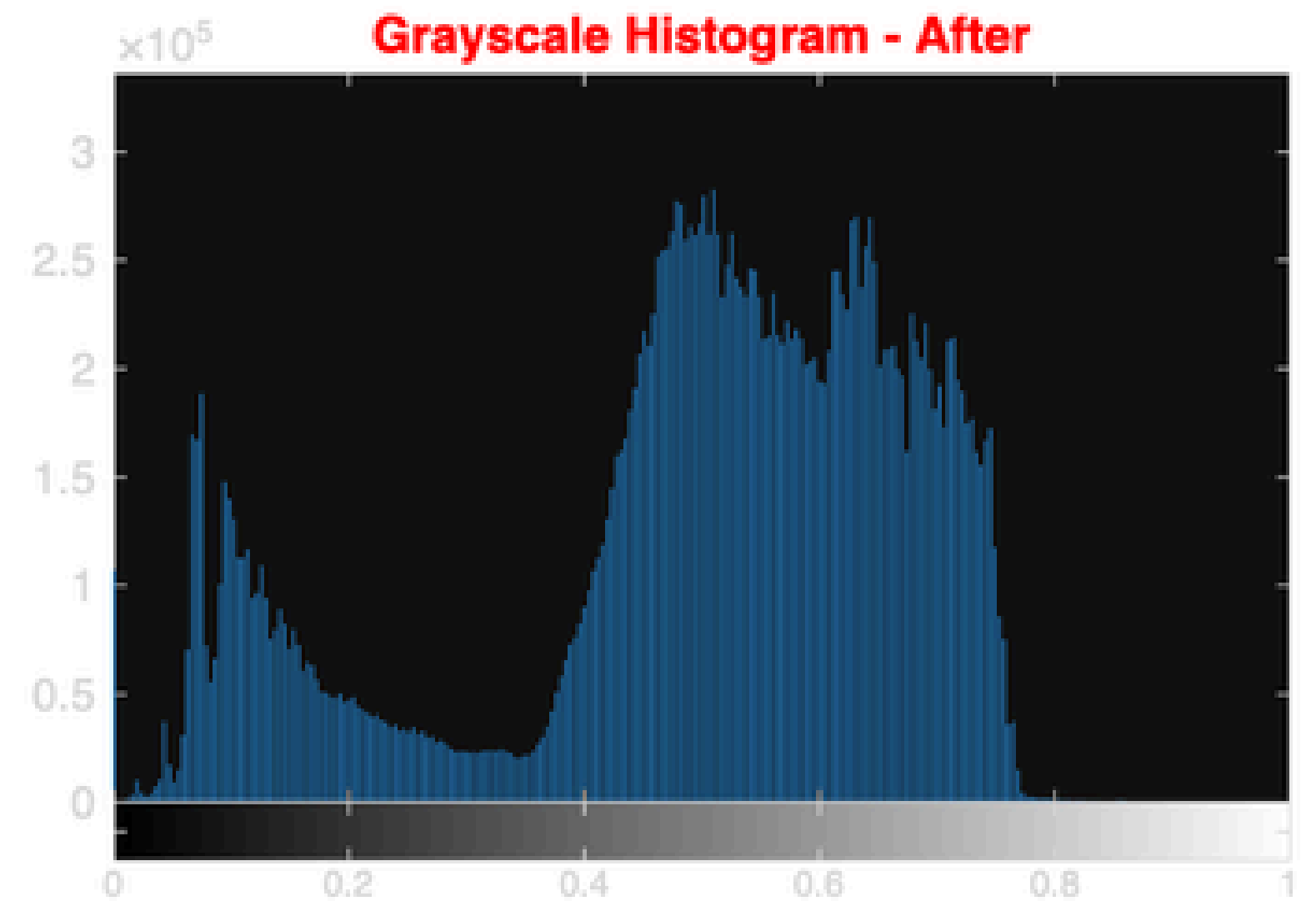
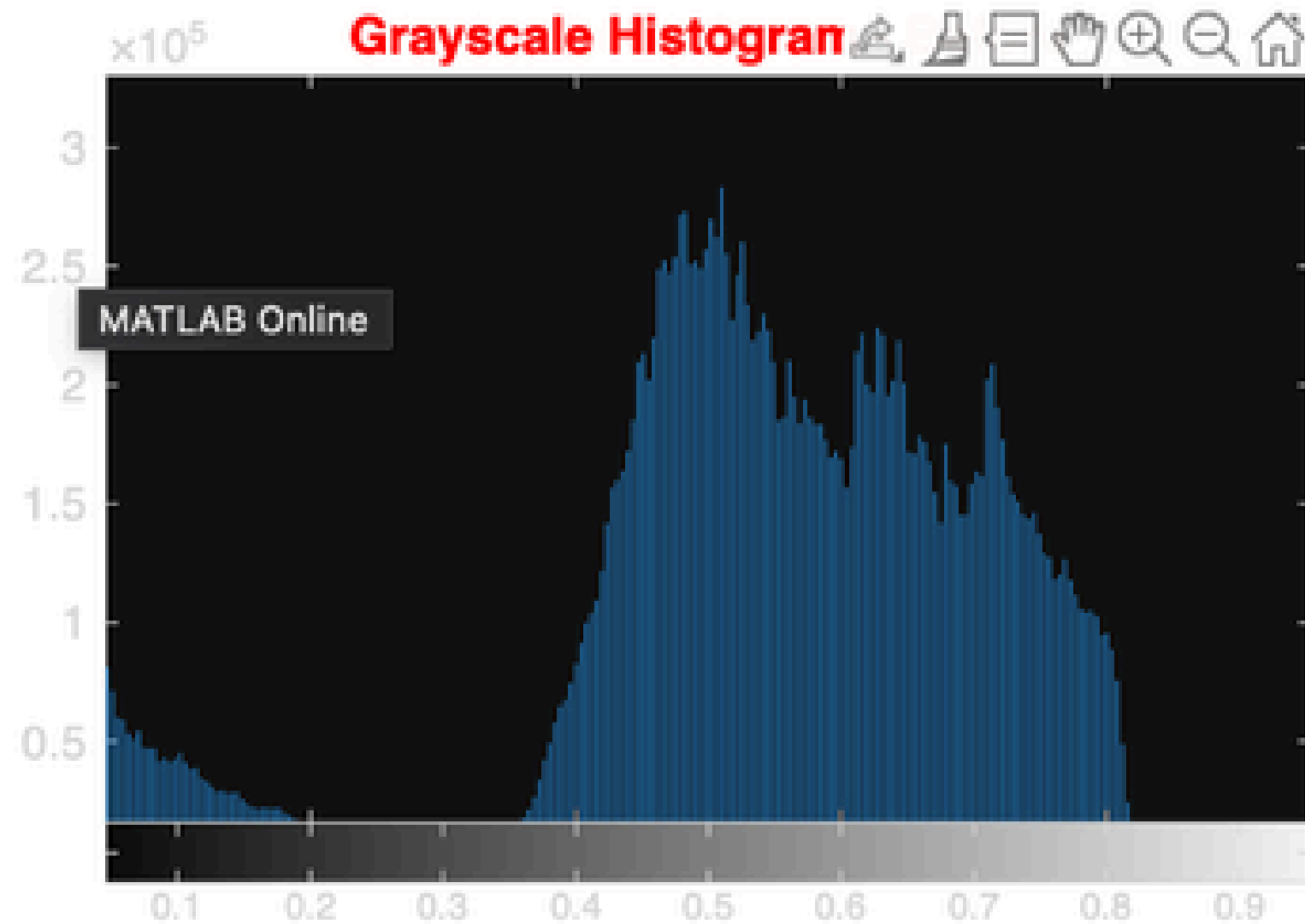
Gaussian Filtering &
Detail Layer Extraction
Detail-Layer Sharpening (Unsharp Mask
Concept)

CONCEPT USED(CONTI.)

```
%% 5) Sharpen (same as before)  
Y = rgb2gray(rgb1);  
Yb = imgaussfilt(Y,1.0);  
detail = Y - Yb;  
  
amount = 0.60;  
rgbOut = rgb1;  
for c = 1:3  
    rgbOut(:,:,c) = min(max(rgbOut(:,:,c) + amount*detail,0),1);  
end
```

Gaussian Filtering

HISTOGRAM COMPARISON



BEFORE

and

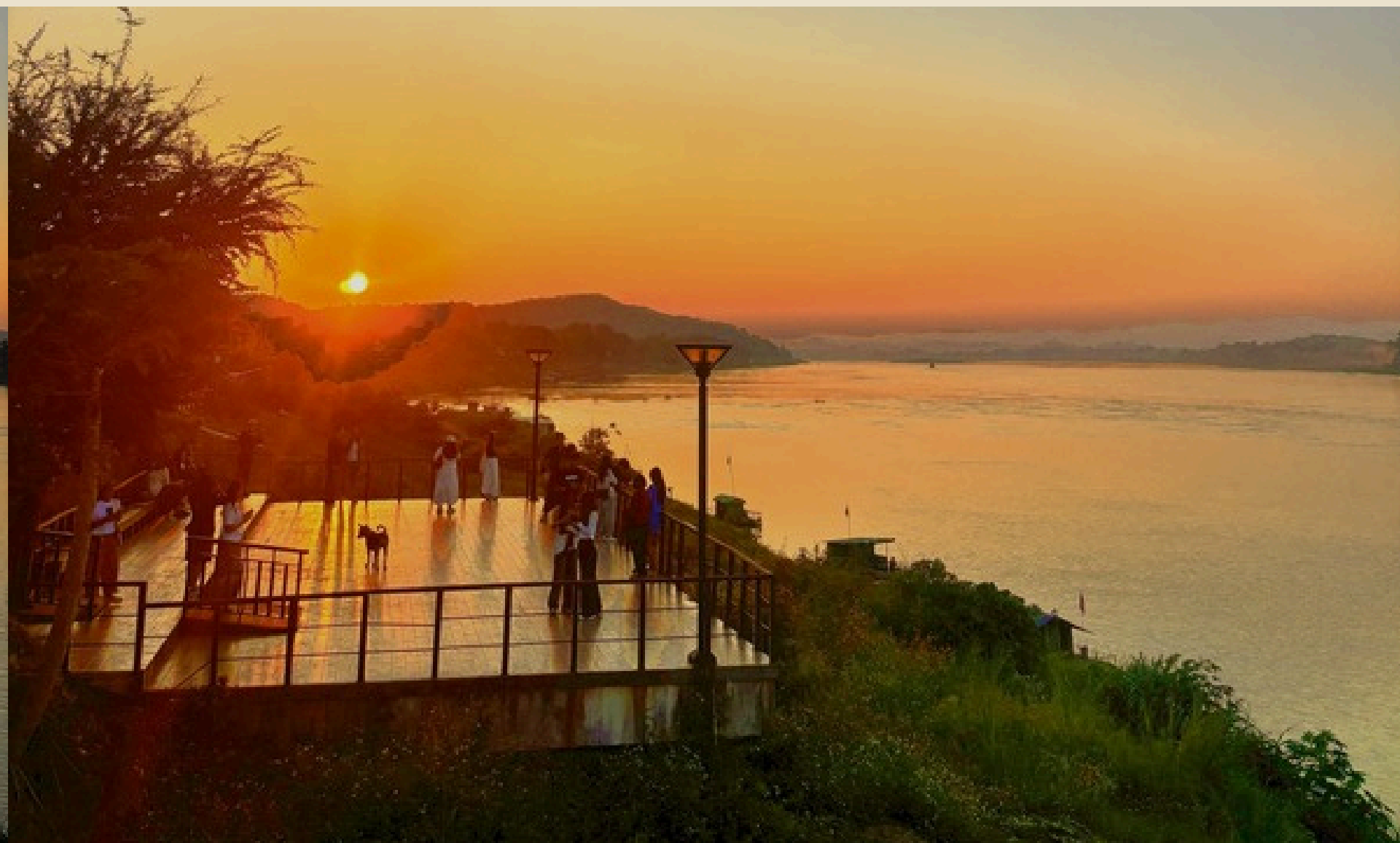
AFTER



BEFORE



AFTER



SUMMARY

- Enhanced shadow visibility using gamma correction
- Controlled sun brightness through highlight compression
- Preserved natural colors using HSV transformation
- Improved clarity using Gaussian filtering and detail sharpening

THANK YOU

